The Cross-Domain Map

The dictionary table, extended, and the words that carry two objects at once

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This is the question I set myself, and it is still the shortest statement of what the file is for.

> *"Try to match terminologies about opponent modeling, transfer learning, cyber environments across cybersecurity, AI, and human factors community."*
>
> q_n_a.tex, open question 39

Three rows of it already sit on the dictionary page: opponent modeling, threat modeling, adversary emulation, against five columns. The columns stay. What follows adds the rest of the rows, in five families.

A row is one word as one community uses it, and the four things saying it commits me to. Words that name the same object from three sides sit next to each other, so that what the row shows is the difference that survives when I cross from one community into the next.

Read Section 5 first if you are short of time. The rows are the easy half. The words that carry two objects at once are the ones that stop a conversation, and I have been on the wrong side of every one of them.

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1 The table

Concept	What is represented?	Why represent it?	Relationship to reality	Output
Reading the other agent				
Opponent modeling	decision process, strategy, policy, beliefs/type	predict/respond strategically	infer/learn a model of an interacting agent	model used for decision-making
Threat modeling	threats, adversaries, capabilities, attack paths, assets/exposure	identify and reason about possible harm	characterize a threat space, often prospectively	structured account of threats/risk
Adversary emulation	observable behaviors/TTPs of an adversary	test defenses against realistic behavior	deliberately reproduce selected real-world behavior	enacted adversarial behavior
Agent modeling	another agent's kind, cooperative or adversarial	coordinate with it or compete with it	recover a hidden type from behavior while the other party adapts	a type estimate carried into the next round
Attacker profile	a stipulated attacker, a script or a parameter set	hold the opponent fixed so a defender result is attributable	set by the experimenter and varied on a known schedule	a runnable opponent
Cognitive model	how an attacker values gains, losses and risk	face the defender with a realistic opponent rather than an idealized one	put the deviation in the reward and let the trajectory follow	a generative attacker whose paths can be run
Opponent embedding	attacker type and recent attacker behavior	keep the adversary's contribution separable from the network's	inferred only from defender-observable signals	a latent the reward predictor reads
Categorization	a small set of partner kinds, relative to the group	carry many relationships on little computation	compress observed partners against the group mean, not a fixed threshold	a classification that moves when the room moves
Moving a policy				
Transfer learning	competence held in a policy or a representation	spend it where it was not trained	relate two environments that describe the world differently	a policy executable in the target
Sim2sim	the gap between two fully specified simulators	measure the gap without an operational network to measure against	both sides are known, so the abstraction between them can be checked	a drop that can be attributed
Sim2real	the gap between a simulator and a deployment	put the policy where the network is	the target is known only in part	a policy that survives live services, or does not
Zero-shot transfer	competence carried with no retraining in the target	test whether the representation did the work	source and target share an interface and nothing else	the same policy, run unchanged
Domain adaptation	the distributional gap left after the schema matches	make one feature carry one number in both	learned from observations of a random policy in each	an encoder a discriminator cannot read the source off
Action translation	what an action does, apart from how it is written	separate the intent from the target's syntax	a shared vocabulary that wrapper logic resolves	one action name per simulator

Concept	What is represented?	Why represent it?	Relationship to reality	Output
State alignment	each observation as kill-chain stage per host	give two environments one schema	computed deterministically, nothing learned	a fixed-size vector in both
Where the agent runs				
Simulator	an abstraction of a network and of acting on it	afford the millions of steps learning wants	models the system rather than executing it	cheap steps, and an unmeasured gap
Emulator	provisioned infrastructure running real services	measure a policy against real tool execution	executes the behavior rather than modeling it	slow steps, honest ones
Benchmark	one frozen configuration and a score	make results comparable across groups	fixes both what was modeled and what was omitted	a number, and the conditions it was taken under
Human-in-the-loop simulation	a configured network with people inside it	watch what an attacker does when part of it is fake	real play, at human speed	observed human behavior, not trajectories
Realism dimension	what an environment provides, models, or emits	decide suitability per claim rather than per environment	derived from what each technique requires in ATT&CK and D3FEND	a fit score for one environment and one objective
Assembling a population				
Strategy population	the strategies that enter the empirical game	represent the full game under a computational budget	every payoff entry is estimated by simulation	an empirical game, and the regret left in it
Population-based training	a set of learners and their hyperparameters	spend a fixed budget on a schedule rather than on one setting	the population is the training run, not a claim about anyone	one model, and the schedule that made it
Attacker population	a spread of stipulated adversaries	face the defender with a distribution instead of one opponent	profiles fixed within a batch, varied across batches	a defender measured against the spread
Reference group	the others a partner is judged against	fix what counts as cooperative here	the group is the standard, so identical evidence reads differently elsewhere	a relative classification
The unseen				
Zero-day	a vulnerability in use before its patch exists	prepare for what no signature covers	counts the days the vendor has had, not what the defender has seen	an incident with no rule waiting for it
Zero-shot	a target with no example and no retraining	test what the representation alone carries	counts what the learner was handed	a policy run unchanged
Few-shot	a target with a handful of episodes	ask what the smallest useful adaptation is	counts the same thing at a different value	an adapted policy, and its budget
Out of distribution	states outside the model's support	bound the behavior where the model is wrong	names a region rather than counting anything	a bound, or a detector
Zero-day condition	a team facing an attack with no prior signature	manipulate what the team has to work from	stipulated in the design, not encountered	a contrast between conditions

2 What separates a family

The dictionary page gives five dimensions and says they are examples rather than a closed list.

1. **Object**: actor, behavior, attack possibilities, environment
2. **Epistemic operation**: describe, infer, predict, simulate/enact
3. **Specificity**: generic adversary, adversary class, particular actor
4. **Purpose**: strategic response, risk analysis, defensive evaluation
5. **Fidelity**: abstract representation, executable reproduction

Each family turns on one of them, and knowing which one is what makes the family worth having as a family.

Reading the other agent turns on epistemic operation. Threat modeling describes, opponent modeling and agent modeling infer, adversary emulation and the cognitive model enact. Specificity does the second cut: threat modeling is at its most useful over an adversary class, adversary emulation over a particular actor.

Moving a policy turns on object. Every row moves something different. Transfer learning moves a policy, domain adaptation moves the numbers on an observation, action translation moves an action, state alignment moves the schema underneath both. Naming the family by what it moves is what keeps the four from being read as four names for one procedure, which is how they get read.

Where the agent runs turns on fidelity, and it is the only family that lies along a single dimension. Which is why six environments can be put on three axes and still come out unordered.

> *"Notice that fidelity and tractability pull against each other. The environments closest to a real system are the ones you can least afford to train in; the ones you can train in fastest are the ones whose results transfer least."*
>
> Cyber Environments & Benchmarks, overview

Assembling a population turns on purpose, and object separates nothing at all. All four rows hold a set of agents. PSRO's set is what you play against, the population-based training set is what you train with, the attacker population is what you are trained against, and the reference group is what you are judged against. One object, four jobs.

The unseen turns on none of the five. That is the finding, not a gap in my placement. The other four families all represent something about an adversary or an environment, and the five dimensions were built for exactly that. This family represents the defender's own evidence: what is available, and how little of it. The Object dimension has no value to give it.

So the map takes a sixth dimension, in the same spirit as the first five.

6. **Evidence**: what is counted, and whose it is. Vendor remediation days, examples handed to a learner, the support of a fitted model, what a team knows before the episode starts.

Placed on that dimension the family separates cleanly, and my own question about it stops being a riddle. More on that in Section 5.

3 Two that move together

A concept in this scheme is a region per dimension plus the correlations between them, and a table of independent columns cannot hold a correlation. Two of them are live here and both are worth writing under the table rather than leaving to the geometry.

Fidelity against tractability. The environments family carries a negative correlation strong enough that I have never seen it broken: cost per step rises with everything that makes a result trustworthy. Tractability is not one of the five dimensions, and it should be. An environment that scores well on fidelity and cannot be run is not a position on the trade, it is out of the family.

Specificity with fidelity. In the reading family these two move together in every one of the three original rows. Threat modeling is generic and abstract, adversary emulation is particular and executable, opponent modeling sits between them on both. Two columns that agree everywhere may be one column wearing two names, so the check is to find a term at an intermediate point on one and see whether the other is forced.

The attacker profile is that test. A scripted B-line agent is entirely particular, it is one named opponent with a fixed route, and it is entirely abstract, since no real tool executes and no service is touched. Specificity is high, fidelity is low, and the correlation breaks. Both columns survive, and the profile row is the one that proved it.

4 Three names, one object

Four of the five families have a word in all three communities. One does not.

Reading the other agent. Cybersecurity says threat modeling, adversary emulation, attacker profile. AI says opponent modeling, agent modeling, type inference, opponent embedding. Human factors says cognitive model, categorization, contrast. The three communities differ on where the model comes from: security stipulates it, AI infers it from play, human factors fits it to observed people.

Where the agent runs. Cybersecurity says range, testbed, emulation. AI says environment, simulator, benchmark. Human factors says human-in-the-loop simulation, and puts participants in it. The three differ on what a step costs and on who takes it.

Assembling a population. Cybersecurity says a set of attacker profiles. AI says strategy population in the game-theoretic sense and population of learners in the optimization sense, which are not each other. Human factors says reference group, and means the population that sets the standard rather than the one being trained.

The unseen. Cybersecurity says zero-day. AI says zero-shot, few-shot, out of distribution. Human factors says the zero-day condition, an experimental manipulation in which a team has no prior signature to work from. Three vocabularies, three counters, and not one of them counts the same quantity.

Moving a policy is the one with nothing in the third column. Security and AI both have the word and mean roughly compatible things, and nothing on my pages says what a human factors researcher would call it. The question it leaves open is worth asking on its own: when an analyst's competence in one network shows up in another, what is it that moved, and would anyone call that transfer?

5 Words that collide

Each of these is one word carrying more than one object, in my own files or across the communities I work with. I am recording them as collisions rather than picking a winner, because in every case both senses are doing real work somewhere.

Environment, and whether the opponent is in it

In reinforcement learning the environment is everything outside the agent, which puts the adversary inside it. FOE-Dreamer takes the opposite position and gives the opponent a latent of its own, so that "an error there is contained rather than smeared across the whole representation."

Learn Structure then puts the boundary back where dependence is weak: a region with no strategic content "collapses into an ordinary single-agent decision," which is to say the opponent rejoins the environment there.

The word marks a boundary that both projects deliberately move. Where the boundary goes is the research, so the word cannot be fixed without giving up the question.

Model, six objects

Threat model, world model, mental model, opponent model, cognitive model, model-based reinforcement learning. Two of them, threat model and mental model, are structured accounts a person writes down. Three of them are fitted objects a machine holds. One, the cognitive model, is a fitted object standing in for a person.

In a room with all three communities the sentence "the model was wrong" is not one claim. It is at least three, and the repair for each is different.

Simulation

The FOE-Dreamer abstract says the defender trains "without a simulator stage."

> **A1** *"the authors mention that they dont use simulation. However, one could say that the world model generates simulations."*
>
> Review 724A, comment A1

Both are right. In the security sense a simulation is a modeled stand-in for a network, and there is none. In the model-based reinforcement learning sense a simulation is an imagined rollout, and the world model produces them by design. The sentence needs the qualifier every time it is written.

Transfer, and what is moving

Sim2Sim before Sim2Real moves an offensive policy out of the environment it learned in. Training in Realistic Environments keeps a defensive policy where it is and moves nothing. Both pages are about the sim-to-real gap, and the word transfer carries a different object in each.

The difference is not cosmetic: one is measured by a drop after the move, the other by a training budget before there was ever a move to make.

Population, three jobs

In PSRO the population is the set of strategies you play against, and its job is coverage: "assembling a strategy population that represents the full game under a computational budget." In population-based training the population is a set of learners run together and mutated, and its job is optimization, not opponents.
In the categorization work the population is the group a partner is scored against, and its job is the standard: "the same behaviour lands in different categories depending on who else is in the room."
Play against, train with, be judged by. Ask which one is meant before agreeing that more population is better.

Zero, two counters

The zero in zero-day counts the days the vendor has had to fix the flaw. The zero in zero-shot counts the examples the learner was given. Neither counts what the defender has seen, which is what my own question was actually asking about.

> *"What is zero-day? Is the defense against zero-days a zero-shot or few-shot challenge?"*
>
> Headspace

It is both, at two grains, and the grain is the answer. At the level of the exploit signature it is zero-shot, because there is no prior instance and nothing to match against. At the level of behavior it is few-shot, because the kill chain recurs even when the vulnerability does not. That is not a compromise between the two, it is the reason my transfer work aligns states "by kill-chain stage per host" rather than by anything the exploit touches. A defense built on the second grain has examples. A defense built on the first never will.

Profile, three objects

A scripted attacker profile is a route through a network: B-line takes a prepared sequence, Meander scans breadth first. A prospect-theory profile is three numbers, the curvature on gains and losses and the loss-aversion multiplier. MDP profiling of a cyber environment is a characterization of the environment itself, with no attacker in it at all.
Route, parameters, environment. The word is doing nothing but signaling that something has been held fixed.

Realistic

In common security usage realistic is a property of an environment, and a claim to it is a claim about the whole environment. The realism work makes it a property of a pair.

> *"Suitability is a property of the* pair*, not of the environment."*
>
> When We Say "A Realistic Cyber Environment"

GOAD is suitable for credential-based privilege escalation at a fit of 0.88 and not suitable for targeted data exfiltration at 0.39, unchanged in between. Anyone using the first sense will read the second as hedging, and anyone using the second will read the first as an unsupported claim. This one is worth correcting in the room rather than letting it pass.

6 Sources

The three original rows, the five dimensions, and the description line come from the Cyber-Human-AI Dictionary page. The rows added here come from the pages where I already use the term: the four cluster overviews and their project pages, the environments blog in full, the Learn Structure notes for the manipulated-opponent schedule, and open question 39.

Two facts came from outside my own files. Population-based training optimizes a population of models together with their hyperparameters under a fixed budget, copying weights from the best performers and mutating the rest, from Jaderberg et al., arXiv 1711.09846. The zero in zero-day counts the days the vendor has had to address the flaw, which is the sense the term settled into after starting out as the day a piece of software was released.

Quotations are exact. Where an original punctuates with a long dash, the dash is set here as a comma or a colon.